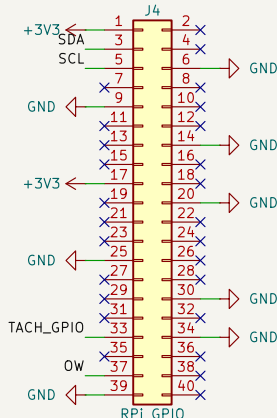
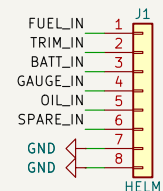


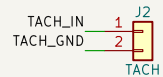
RASPBERRY PI HEADER



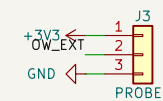
HELM: gauge taps and battery



TACH: Delco EST (ignition side)

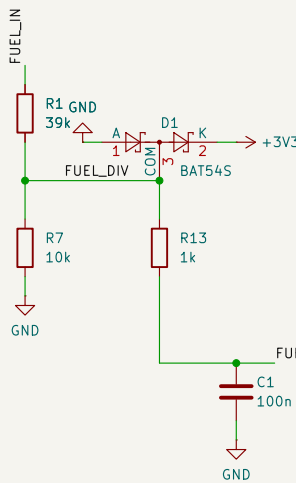


TEMPERATURE PROBES

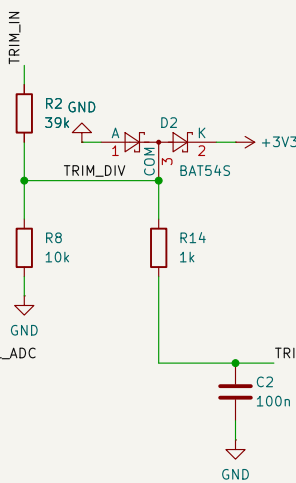


Each gauge tap: the S (or I) terminal -> 10k in heatshrink AT THE GAUGE -> J1 -> 39k -> node (10k to GND, BAT54S clamp to GND/3V3) -> 1k -> ADC, with 100n at the ADC pin. $V_{adc} = V_{tap} \times 10/59$ (0-16 V -> 0-2.7 V). Battery: 47k on board, no remote resistor, x 10/57. Tach, Delco EST ignition (the newer coil: 12 V in, tach out): J2-1 to the gray tach wire, J2-2 to engine ground. The original tach stays on the gray wire. Takes a 12 V tach signal (~1 mA) or a coil-negative wire (spikes to a few hundred volts): three 3.3k 1206 in series share the spike: 1N4148W and 10n across the LED. One rising edge per ignition cycle on GPIO13; 2 per revolution on the 3.0 4-cylinder (BOAT_TACH_PPR=2). Keep every TACH_* net (the ignition side) 3 mm from all other copper: see sensor-board.kicad_dru. Software: BOAT_SENSORS=real BOAT_SENDER_WIRING=tap BOAT_TACH_PULL=none BOAT_TACH_GPIO=13 BOAT_OIL_SENDER=true; dtoverlay=w1-gpio,gpiopin=26. Channel map: fuel 0, trim 1, battery 2, gauge supply 3, oil 4 -- see SENSOR_BOARD.md.

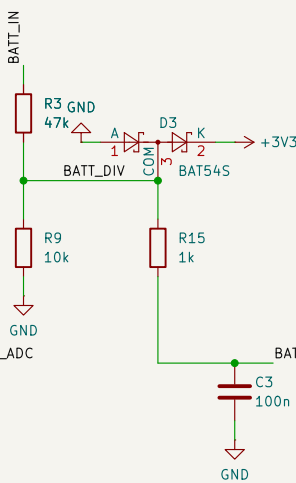
FUEL: J1.1 -> U1 AIN0



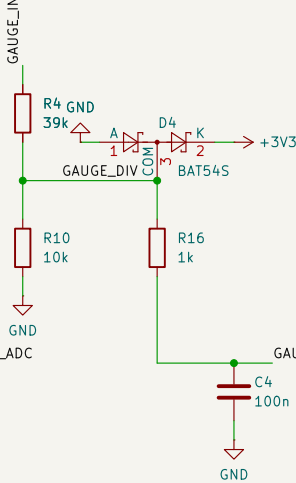
TRIM: J1.2 -> U1 AIN1



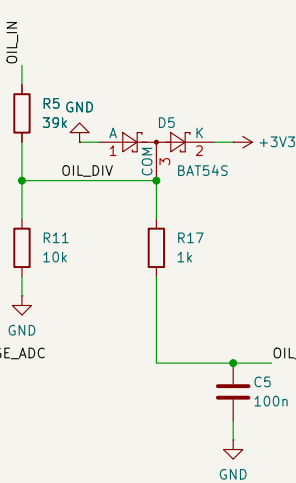
BATT: J1.3 -> U1 AIN2



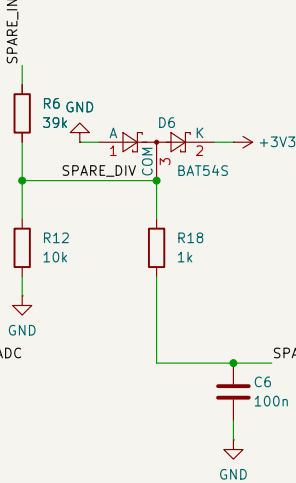
GAUGE: J1.4 -> U1 AIN3



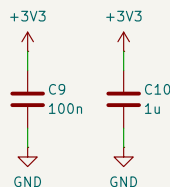
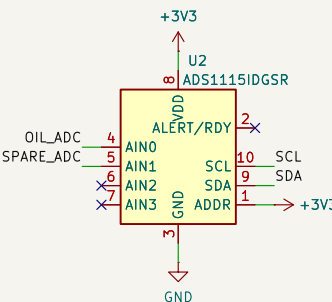
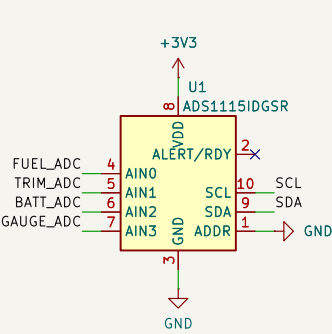
OIL: J1.5 -> U2 AIN0



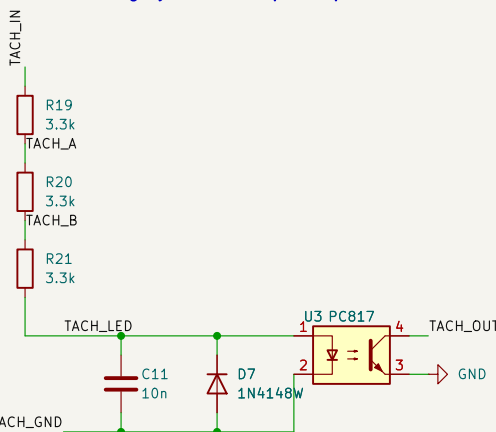
SPARE: J1.6 -> U2 AIN1



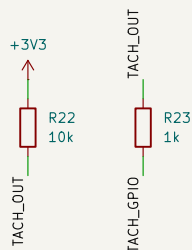
CONVERTERS



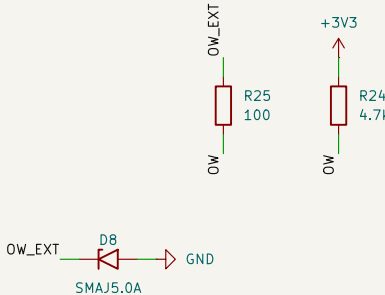
TACH: Delco EST gray wire -> optocoupler



TACH OUTPUT



1-WIRE PROBES



MOUNTING / POWER FLAGS



Generated from design.py by schematic.py; edit those, not this file
Passive taps on the existing analog gauges -- they stay connected and working

Sheet: / File: sensor-board.kicad_sch		
Title: Boat MFD sensor board		
Size: A3	Date: 2026-09-22	Rev: 1.0
KiCad E.D.A. 10.0.6		Id: 1/1